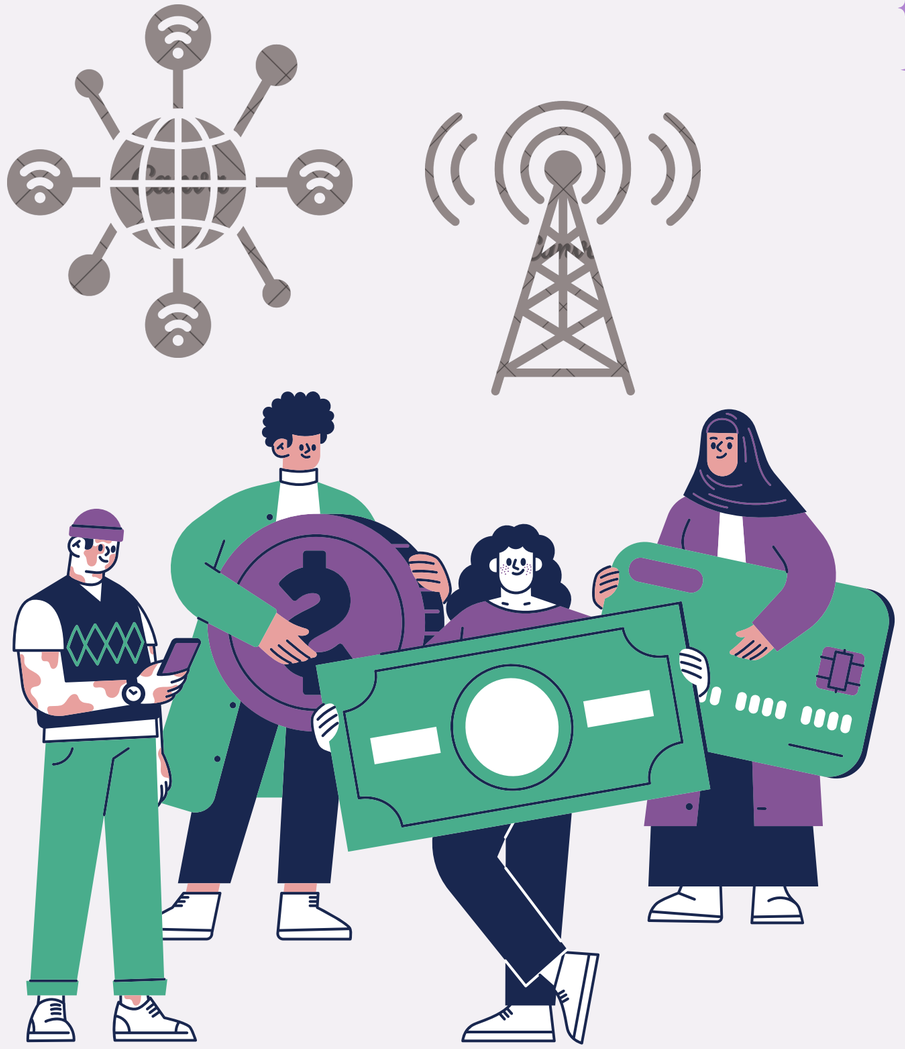
TELCO CUSTOMER CHURN ANALYSIS (SQL CASE STUDY)

USING MICROSOFT SQL



PRESENTED BY

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SQL Server | Data Analytics Portfolio Project

BUSINESS PROBLEMS $ OBJECTIVE

BUSINESS OBJECTIVES

- ASSESS THE OVERALL CHURN RATE AND HIGH RISK SEGMENT
- EVALUATE REVENUE IMPACT (ARPU & REVENUE AT RISK)
- IDENTIFY HIGH-RISK CUSTOMER SEGMENTS
- DETERMINE BEHAVIORAL AND PRICING DRIVERS OF CHURN
- RECOMMEND DATA-DRIVEN RETENTION STRATEGIES

BUSINESS PROBLEMS

THE COMPANY IS EXPERIENCING CUSTOMER CHURN AND WANTS TO UNDERSTAND:

•WHAT IS THE OVERALL CHURN RATE?

•DOES GENDER CATEGORY, SUBSCRIPTION MODEL, BILLING BEHAVIOR AND CUSTOMER LIFECYCLE INFLUENCE CHURN?

•WHICH CUSTOMER SEGMENTS ARE MOST AT RISK

•HOW MUCH REVENUE IS AT RISK AND WHAT IS DRIVING THE ARPU

ABOUT DATASET

THE DATASET WAS EXTRACTED FROM KAGGLE AND IT CONTAINS A TOTAL CUSTOMERS OF 7,043 AND 21 COLUMN FEATURES.

THE KEY VARIABLES ARE

- Churn
- PaymentMethod
- MonthlyCharges
- TotalCharges
- Contract
- Tenure

DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn	ChurnFlag	tenure_group
No	No	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No	0	0 - 12 months
Yes	No	No	No	One year	No	Mailed check	56.95	1889.50	No	0	24 - 48months
No	No	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes	1	0 - 12 months
Yes	Yes	No	No	One year	No	Bank transfer	42.30	1840.75	No	0	48+ months
No	No	No	No	Month-to-month	Yes	Electronic check	70.70	151.65	Yes	1	0 - 12 months
Yes	No	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.50	Yes	1	0 - 12 months
No	No	Yes	No	Month-to-month	Yes	Credit card	89.10	1949.40	No	0	12 - 24 months
No	No	No	No	Month-to-month	No	Mailed check	29.75	301.90	No	0	0 - 12 months
Yes	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	104.80	3046.05	Yes	1	24 - 48months
No	No	No	No	One year	No	Bank transfer	56.15	3487.95	No	0	48+ months
No	No	No	No	Month-to-month	Yes	Mailed check	49.95	587.45	No	0	12 - 24 months
No internet ser...	No internet...	No internet...	No internet ser...	Two year	No	Credit card	18.95	326.80	No	0	12 - 24 months
Yes	No	Yes	Yes	One year	No	Credit card	100.35	5681.10	No	0	48+ months
Yes	No	Yes	Yes	Month-to-month	Yes	Bank transfer	103.70	5036.30	Yes	1	48+ months
Yes	Yes	Yes	Yes	Month-to-month	Yes	Electronic check	105.50	2686.05	No	0	24 - 48months
Yes	Yes	Yes	Yes	Two year	No	Credit card	113.25	7895.15	No	0	48+ months
No internet ser...	No internet...	No internet...	No internet ser...	One year	No	Mailed check	20.65	1022.95	No	0	48+ months
Yes	No	Yes	Yes	Two year	No	Bank transfer	106.70	7382.25	No	0	48+ months
Yes	Yes	No	No	Month-to-month	No	Credit card	55.20	528.35	Yes	1	0 - 12 months
Yes	No	No	Yes	Month-to-month	Yes	Electronic check	90.05	1862.90	No	0	12 - 24 months
Yes	No	No	Yes	Month-to-month	Yes	Electronic check	39.65	39.65	Yes	1	0 - 12 months
No internet ser...	No internet...	No internet...	No internet ser...	One year	No	Bank transfer	19.80	202.25	No	0	0 - 12 months
No internet ser...	No internet...	No internet...	No internet ser...	Month-to-month	No	Mailed check	20.15	20.15	Yes	1	0 - 12 months

7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No
6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No
9763-GRSKD	Male	0	Yes	Yes	13	Yes	No	DSL	Yes	No	No
7469-LKBCI	Male	0	No	No	16	Yes	No	No	No internet s...	No internet ...	No internet ser...
8091-TTVAX	Male	0	Yes	No	58	Yes	Yes	Fiber optic	No	No	Yes
0280-XJGEX	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes
5129-JLPIS	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes
3655-SNQYZ	Female	0	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes
8191-XWSZG	Female	0	No	No	52	Yes	No	No	No internet s...	No internet ...	No internet ser...
9959-WOFKT	Male	0	No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes
4190-MFLUW	Female	0	Yes	Yes	10	Yes	No	DSL	No	No	Yes
4183-MYFRB	Female	0	No	No	21	Yes	No	Fiber optic	No	Yes	Yes
8779-QRDMV	Male	1	No	No	1	No	No phone service	DSL	No	No	Yes
1680-VDC...	Male	0	Yes	No	12	Yes	No	No	No internet s...	No internet ...	No internet ser...
1066-JKSGK	Male	0	No	No	1	Yes	No	No	No internet s...	No internet ...	No internet ser...

OnlineBackup	DeviceProtection
Yes	No
No	Yes
Yes	No

DATA PREPARATION IN SQL

UPDATING THE ACCURATE DATATYPE

```sql
Alter Table dbo.TelcoCustomerChurn
Alter Column TotalCharges Decimal(10,2)

Alter Table dbo.TelcoCustomerChurn
Alter Column MonthlyCharges Decimal(10,2)


Alter Table dbo.TelcoCustomerChurn
Alter Column tenure int
```

CHECKING FOR DUPLICATES AND TRIMMING SPACES

```sql
SELECT customerID, COUNT(*) AS duplicate_count
FROM dbo.TelcoCustomerChurn
GROUP BY customerID
HAVING COUNT(*) > 1;

UPDATE dbo.TelcoCustomerChurn
SET
    customerID = LTRIM(RTRIM(customerID)),
    PaymentMethod = LTRIM(RTRIM(PaymentMethod)),
    Contract = LTRIM(RTRIM(Contract));
```

CREATING NUMERICAL CHURN COLUMN

```sql
Alter Table dbo.TelcoCustomerChurn
Add ChurnFlag int

update dbo.TelcoCustomerChurn
set ChurnFlag = CASE When Churn = 'yes' then 1
Else 0
End

Select Churn, ChurnFlag
from dbo.TelcoCustomerChurn
```

CLEANING UNNECESSARY DETAILS

```sql
Update dbo.TelcoCustomerChurn
Set PaymentMethod = Replace(PaymentMethod, '(automatic)','')
where PaymentMethod like '%(automatic)%'
```

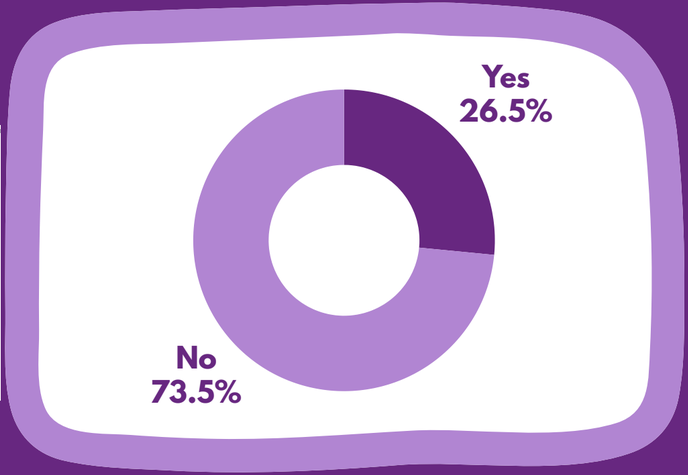
CHURN RATE CALCULATION

OVERALL CHURN RATE

Churn Rate = Churned Customers / Total Customers

```
Select churn,
count(*) As Customers,
Cast(Count(*)*100.0/(Select Count(*)
From dbo.TelcoCustomerChurn) As decimal(10,2)) As Percentage
From dbo.TelcoCustomerChurn
group by churn
Order by count(*);
```

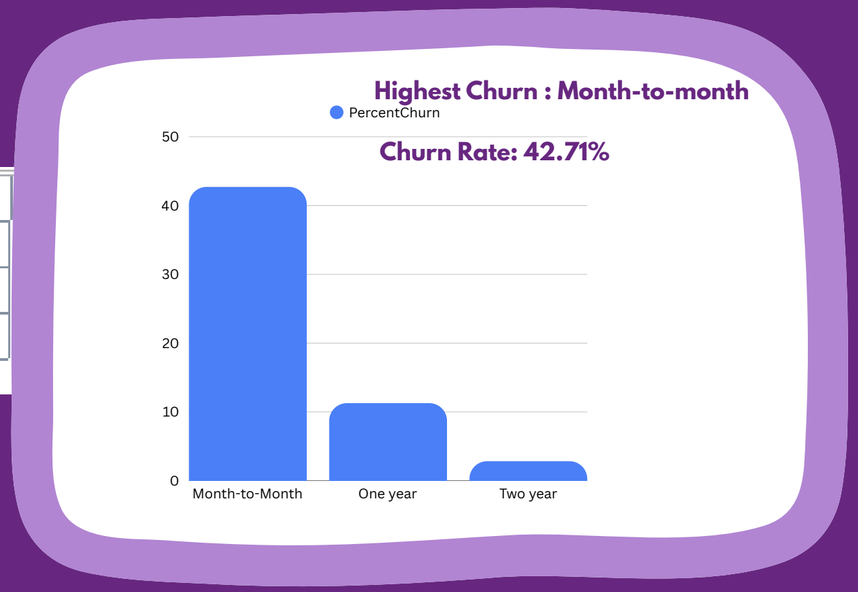
churn	Customers	Percentage
Yes	1869	26.54
No	5174	73.46



CHURN BY CONTRACT

```
SELECT
    Contract,

    COUNT(*) AS Total_Customers,

    SUM(ChurnFlag)
        AS Churned_Customers,

    CAST(
        SUM(ChurnFlag) * 100.0
        / COUNT(*)
        AS DECIMAL(10,2)
    ) AS Churn_Percentage

FROM dbo.TelcoCustomerChurn
GROUP BY Contract
ORDER BY Churn_Percentage DESC;
```

Contract	Total_Customers	Churned_Customers	Churn_Percentage
Month-to-month	3875	1655	42.71
One year	1473	166	11.27
Two year	1695	48	2.83



CHURN RATE CALCULATION

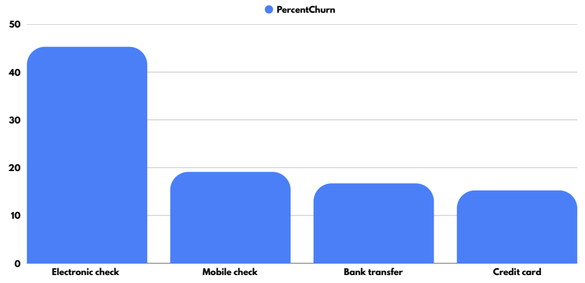
CHURN BY PAYMENTMETHOD

```sql
SELECT
    PaymentMethod,

    COUNT(*) AS Total_Customers,

    SUM(ChurnFlag)
        AS Churned_Customers,

    CAST(
        SUM(ChurnFlag) * 100.0
        / COUNT(*)
        AS DECIMAL(10,2)
    ) AS Churn_Percentage

FROM dbo.TelcoCustomerChurn
GROUP BY PaymentMethod
ORDER BY Churn_Percentage DESC;
```

PaymentMethod	Total_Customers	Churned_Customers	Churn_Percentage
Electronic check	2365	1071	45.29
Mailed check	1612	308	19.11
Bank transfer	1544	258	16.71
Credit card	1522	232	15.24

Highest Churn : Electronic check

Churn Rate: 45.29%



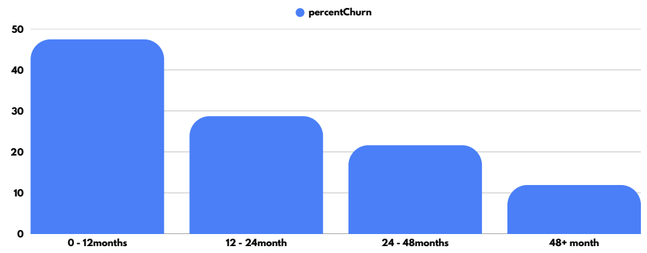
CHURN BY TENURE GROUP

```sql
SELECT
    tenure_group,

    COUNT(*) AS Total_Customers,

    SUM(ChurnFlag)
        AS Churned_Customers,

    CAST(
        SUM(ChurnFlag) * 100.0
        / COUNT(*)
        AS DECIMAL(10,2)
    ) AS Churn_Percentage

FROM dbo.TelcoCustomerChurn
GROUP BY tenure_group
ORDER BY Churn_Percentage DESC;
```

tenure_group	Total_Customers	Churned_Customers	Churn_Percentage
0 - 12 months	2186	1037	47.44
12 - 24 months	1024	294	28.71
24 - 48months	832	180	21.63
48+ months	3001	358	11.93

Highest Churn : 0-12months

Churn Rate: 47.44%

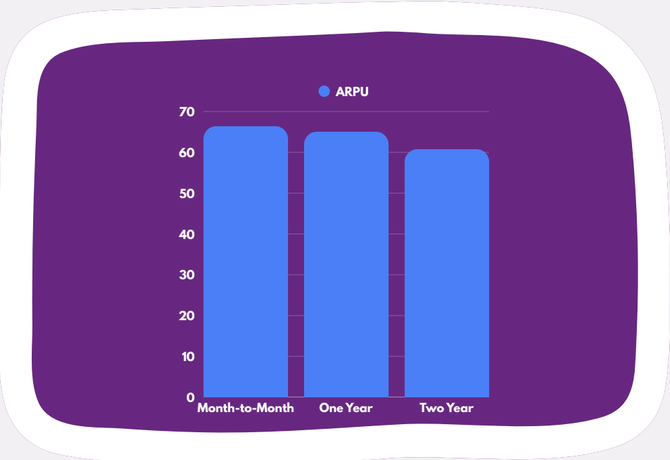


REVENUE DRIVERS

ARPU BY CONTRACT

```
Select Contract,
Count(*) as Customers,
Cast(AVG(MonthlyCharges) As decimal(10,2)) As ARPU,
Cast(Sum(MonthlyCharges) As decimal(10,2)) As Total_Revenue
from dbo.TelcoCustomerChurn
group by Contract
Order by ARPU Desc
```

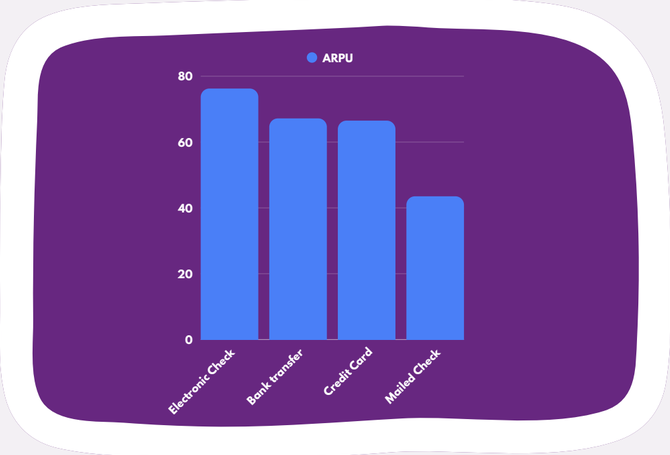
Contract	Customers	ARPU	Total_Revenue
Month-to-month	3875	66.40	257294.15
One year	1473	65.05	95816.60
Two year	1695	60.77	103005.85



ARPU BY PAYMENTMETHOD

```
Select PaymentMethod,
Count(*) as Customers,
Cast(AVG(MonthlyCharges) As decimal(10,2)) As ARPU,
Cast(Sum(MonthlyCharges) As decimal(10,2)) As Total_Revenue
From dbo.TelcoCustomerChurn
group by PaymentMethod
order by ARPU Desc
```

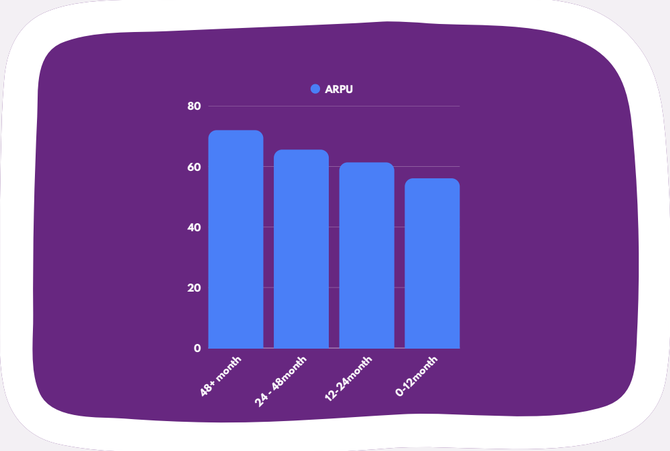
	PaymentMethod	Customers	ARPU	Total_Revenue
1	Electronic check	2365	76.26	180345.00
2	Bank transfer	1544	67.19	103745.45
3	Credit card	1522	66.51	101231.85
4	Mailed check	1612	43.92	70794.30



ARPU BY tenure group

```
Select tenure_group,
Count(*) as Customers,
Cast(AVG(MonthlyCharges) As decimal(10,2)) As ARPU,
Cast(Sum(MonthlyCharges) As decimal(10,2)) As Total_Revenue
From dbo.TelcoCustomerChurn
group by tenure_group
order by ARPU Desc
```

	tenure_group	Customers	ARPU	Total_Revenue
1	48+ months	3001	72.01	216098.20
2	24 - 48months	832	65.58	54558.80
3	12 - 24 months	1024	61.36	62829.85
4	0 - 12 months	2186	56.10	122629.75



CHURN VS NON-CHURN

```sql
SELECT
    Churn,
    COUNT(*) AS Total_Customers,
    CAST(AVG(MonthlyCharges) AS DECIMAL(10,2)) AS ARPU,
    CAST(SUM(MonthlyCharges) AS DECIMAL(10,2)) AS Total_Revenue
FROM dbo.TelcoCustomerChurn
GROUP BY Churn;
```

Churn	Total_Customers	ARPU	Total_Revenue
Yes	1869	74.44	139130.85
No	5174	61.27	316985.75

INSIGHT

- Cusomers that churned, generated a high ARPU compared to Non-Churn Customers
- There is significant impact by the revenue loss
- Retention of customer with high ARPU is crucial

REVENUE AT RISK

```sql
Select
Cast(Sum(MonthlyCharges) As decimal(10,2)) As RevenueAtRisk
From dbo.TelcoCustomerChurn
Where Churn = 'Yes'
```

RevenueAtRisk
139130.85

The revenue at risk from churned customers is $139,130.85

KEY INSIGHT AND RECOMMENDATIONS

KEY INSIGHT

- Month-to-month contract drives the highest ARPU and highest churn rete
- Customers that uses auto-payment drives lower churn rate with a stable revenue
- Longer contract improves revenue pedictability
- High ARPU churnners represent a high revenue risk

RECOMMENDATIONS

- Incentivize longer contract adoption
- Promote auto-payment enrollment
- Target high-ARPU churn-risk customers with retention offers
- Introduce loyalty pricing models

THANK YOU

Data is only powerful when it drives action.
Let’s transform complexity into clarity.